CATCH: Counterfactual Anatomical Tissue Inpainting with a Conditional Haar-Wavelet Diffusion Model

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Abstract. Automated neuroimaging pipelines are typically built for healthy anatomy, so brain tumors that distort surrounding tissue can degrade their reliability. The BraTS local synthesis task addresses this by asking for anatomically plausible, tumor-free (counterfactual) tissue to be synthesized inside a masked region of a brain MRI, while the observed anatomy is left unchanged. We present a 3D wavelet-domain conditional denoising diffusion model for this inpainting task. The diffusion process runs on the eight-channel 3D Haar wavelet representation of the volume, which keeps full-resolution volumetric generation tractable on a single GPU. Our baseline concatenates the noisy target subbands, the voided-image subbands, and a compact signed mask. We also investigate a spatial FiLM variant that uses the voided image and mask to modulate the eight noisy target subbands before the U-Net. The model predicts the clean target directly and restores observed voxels exactly through hard-blend compositing. To match the challenge setting, we exclude tumor-containing cells from reconstruction loss, normalize a mask-weighted term by the active healthy-mask area, and study online augmentation of the artificially removed healthy region. We evaluate with the official healthy-mask SSIM, PSNR, and MSE metrics on a deterministic 100-case hold-out from the BraTS local synthesis training data.

Keywords: MRI inpainting · Brain MRI · Medical image synthesis · Diffusion models · Wavelet transform · BraTS

1 Introduction

Brain tumors distort the surrounding anatomy and violate the tissue-appearance assumptions of many automated neuroimaging pipelines. Tools for registration, brain-region parcellation, and atlas-based analysis are typically developed and validated on healthy anatomy, and their behavior on scans containing large lesions can be unreliable. A practical way to reconcile pathological scans with such tools is to replace the tumor region with a plausible, tumor-free (counterfactual) completion of the underlying healthy tissue, so that downstream analysis can proceed as if the pathology were absent. The BraTS local synthesis task [8] formalizes this as an inpainting problem: given a brain MRI with a masked-out region and the corresponding mask, synthesize anatomically plausible healthy tissue inside the mask while leaving the observed anatomy unchanged.

Denoising diffusion probabilistic models [7] have become the leading approach for high-fidelity image synthesis and inpainting [10,16], and are increasingly used for 3D medical image generation. Applying them directly to full-resolution brain volumes is memory intensive, however, which has motivated diffusion in a compressed representation such as the wavelet domain [6,5]. Building on this line of work, we develop a 3D wavelet-domain conditional diffusion model for the BraTS healthy-tissue inpainting task and adapt both its conditioning and its training objective to the specific structure of the challenge.

We do not claim to introduce wavelet-domain diffusion; rather, we present a careful adaptation and extension of cWDM/WDM-style 3D wavelet diffusion [6,5] to healthy-tissue inpainting, together with a reproducible challenge pipeline. Our contributions are:

- A **3D wavelet-domain conditional DDPM** for BraTS healthy-tissue inpainting. The model runs the diffusion process on the eight-channel 3D Haar wavelet representation of the volume, represents the spatial mask as one signed channel, predicts the clean target directly (predict-x_0), and reconstructs the missing tissue with a hard-blend compositing step that keeps all observed voxels pixel-exact. We use 17-channel concatenation in the final experiments and study an eight-channel spatial input-FiLM formulation during development.
- A **task-specific healthy-tissue training objective**. We supervise the artificially removed healthy region, where ground truth is available, exclude the tumor region from reconstruction, and normalize a soft mask-weighted MSE by its active mass so its coefficient has consistent meaning across mask sizes.
- **Online healthy-mask augmentation and reproducible evaluation**. We compare fixed masks with augmented tumor shapes and a mixture of irregular mask families. We enforce split isolation and no overlap with the real tumor. All variants use a deterministic held-out split and the official healthy-mask SSIM/PSNR/MSE implementation.

2 Related Work

Diffusion models. Denoising diffusion probabilistic models [7] generate data by learning to reverse a gradual noising process, and were shown to rival or surpass GANs on image synthesis [3]. A continuous, score-based formulation unifies these models through stochastic differential equations and introduces the variance-preserving (VP) schedule [18] that we adopt. Subsequent work improved sample quality and likelihood [12] and accelerated sampling with deterministic implicit samplers [17]. To reduce the cost of high-resolution generation, latent diffusion models run the process in a learned compressed space [14]; wavelet-domain diffusion follows the same motivation using a fixed, invertible transform instead of a learned autoencoder.

Diffusion-based inpainting. Diffusion models are well suited to inpainting because the reverse process can be guided by known pixels. RePaint [10] conditions an unconditional model by repeatedly replacing the known region with a correctly noised version of the observed image during sampling, whereas conditional image-to-image approaches such as Palette [16] feed the observed context directly to the network as an additional input. Our model follows the conditional route: the voided volume and mask are supplied directly to the denoiser, and observed voxels are restored exactly by a final compositing step rather than by per-step resampling. In addition to channel concatenation, we study feature-wise linear modulation (FiLM) [13], which applies a learned affine transformation whose parameters are generated from the conditioning input. Unlike the global per-channel parameters in the original FiLM formulation, our conditioning branch predicts spatial parameter maps aligned with the wavelet grid.

3D medical image synthesis and healthy-tissue inpainting. Generating full 3D medical volumes with diffusion is challenging due to memory constraints. Wavelet diffusion models (WDM) [6] address this by denoising in the 3D discrete wavelet domain, enabling high-resolution volumetric synthesis on a single GPU, and their conditional extension (cWDM) [5] adds channel-wise conditioning for cross-modality and inpainting-style tasks. Diffusion models have also been applied specifically to 3D healthy brain tissue inpainting [4] for the BraTS local synthesis task [8], which builds on the multimodal BraTS benchmark and its data collections [11,2,1]. Our approach is a direct adaptation and extension of this wavelet-diffusion line of work: we retain the 3D wavelet-domain conditional diffusion backbone and tailor the conditioning and training objective to the healthy-tissue inpainting setting of the BraTS challenge.

3 Method

We cast healthy-tissue inpainting as conditional 3D generation with a denoising diffusion model [7] operating in a fixed Haar-wavelet representation, following WDM and cWDM [6,5]. The denoiser sees only information available at challenge inference time: the voided T1n image and the full inpainting mask. Training additionally uses the provided healthy and unhealthy sub-masks to determine where the complete T1n image is a valid healthy target.

Figure 1 summarizes the shared conditioning, training, and reverse-sampling paths.

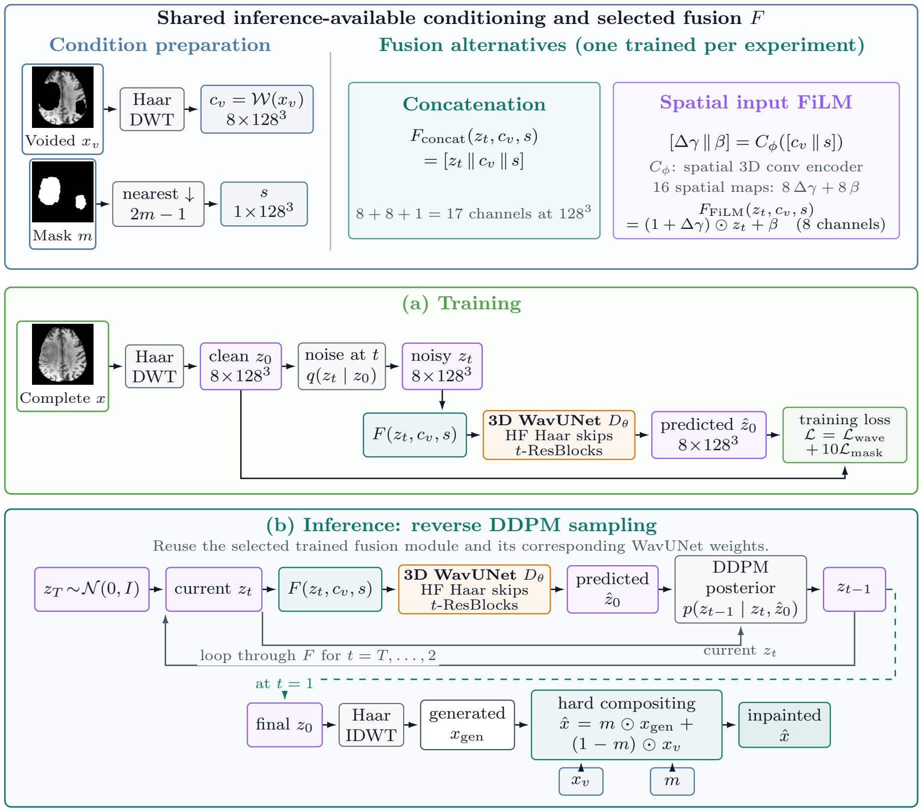

Fig. 1. Overview of CATCH. The voided image and downsampled signed mask are the only inference-time conditions. The framework supports either channel concatenation or spatial input FiLM. During training, the WavUNet predicts clean Haar coefficients and is supervised by the tumor-excluded wavelet loss and healthy-hole-focused mask loss. During inference, the current latent z_t and clean prediction $\hat{z}_0$ enter the DDPM posterior separately, and z_{t-1} returns through the selected fusion module. The final sampled z_0 is decoded into the output volume, after which hard compositing preserves x_v outside m exactly. The thumbnails show a geometry-selected training case on its maximum-mask-area axial slice; they are illustrative and were not selected using reconstruction performance.

3.1 Problem Formulation

Let x_v denote a voided T1n volume and $m \in \{0,1\}^{H\times W\times D}$ the region to synthesize. The desired prediction $\hat{x}$ must contain plausible tumor-free tissue inside m and must equal the observation outside it:

$$\hat{x} \odot (1-m) = x_v \odot (1-m), \tag{1}$$

where $\odot$ is elementwise multiplication. During training, m is partitioned into an artificially removed healthy region m_h and a real tumor region m_u. The complete training image x provides a target in m_h, but still contains pathology in m_u; consequently, m_u must not be treated as healthy supervision. Only x_v and m are used as model inputs, matching the challenge setting.

3.2 Wavelet-Domain Diffusion

BraTS volumes have native size $240 \times 240 \times 155$. We reorient them to canonical RAS orientation and center-pad them to 256^3 without resampling. A single-level 3D Haar discrete wavelet transform (DWT), denoted $\mathcal{W}$, decomposes each volume into the eight subbands $\{\mathrm{LLL}, \mathrm{LLH}, \mathrm{LHL}, \mathrm{LHH}, \mathrm{HLL}, \mathrm{HLH}, \mathrm{HHL}, \mathrm{HHH}\}$, each of size 128^3. We stack the subbands as channels and divide the LLL channel by three to reduce its dynamic-range imbalance; inversion multiplies that channel by three before the inverse DWT. Thus a 256^3 volume is represented by an 8×128^3 tensor without a learned autoencoder.

Let $z_0 = \mathcal{W}(x)$ be the scaled wavelet representation of a complete training image. The $T = 1000$ step forward process is

$$q(z_t \mid z_0) = \mathcal{N}\big(z_t; \sqrt{\bar{\alpha}_t}\, z_0, (1 - \bar{\alpha}_t)\mathbf{I}\big)\,, \tag{2}$$

with a variance-preserving SDE schedule [18] using $\beta_{\min} = 0.1$ and $\beta_{\max} = 20$. The network directly predicts the clean coefficients $\hat{z}_0$ (predict-x_0) and uses a fixed-large reverse variance.

3.3 Spatial Conditioning

The voided image is transformed into eight scaled subbands $c_v = \mathcal{W}(x_v)$. We downsample the binary mask once by nearest-neighbor interpolation to the 128^3 wavelet grid and encode it as $s = 2\,\mathrm{NN}(m) - 1 \in \{-1, 1\}^{128^3}$. The full-resolution binary mask remains separate and is used for the loss and final compositing; the signed tensor s is a model input only.

Our concatenation baseline receives

$$\hat{z}_0 = D_\theta([z_t \,\|\, c_v \,\|\, s], t)\,, \tag{3}$$

where $[\cdot\|\cdot]$ denotes channel concatenation. This produces a 17-channel U-Net input: eight noisy target channels, eight voided-image channels, and one signed mask channel.

We additionally implement spatial input FiLM [13] as an alternative fusion mechanism. A conditioning network F_ϕ receives the nine-channel tensor $[c_v\|s]$ and predicts eight spatial scale residuals $\Delta\gamma$ and eight spatial biases β:

$$[\Delta\gamma \,\|\, \beta] = F_\phi([c_v \,\|\, s]), \tag{4}$$

$$\tilde{z}_t = (1 + \Delta\gamma) \odot z_t + \beta, \tag{5}$$

$$\hat{z}_0 = D_\theta(\tilde{z}_t, t). \tag{6}$$

F_ϕ consists of a $3 \times 3 \times 3$ convolution from 9 to 32 channels, GroupNorm and SiLU, a second $3\times3\times3$ convolution from 32 to 32 channels with GroupNorm and

SiLU, and a $1 \times 1 \times 1$ convolution from 32 to 16 channels. The final convolution is zero-initialized, so FiLM begins as the identity mapping. This variant passes only the eight modulated target subbands to the U-Net. At inference, F_ϕ is evaluated once per case and its spatial maps are reused for the entire reverse trajectory. This input modulation is independent of the existing timestep-derived scale-and-shift modulation inside the residual blocks.

3.4 Wavelet U-Net

D_θ is a fully 3D wavelet U-Net [15,6] with base width 64, channel multipliers $(1, 2, 2, 4, 4, 4)$, and two residual blocks per level. It uses GroupNorm with 32 groups and SiLU activations. A sinusoidal timestep embedding followed by a two-layer MLP supplies scale and shift to every residual block. Haar-wavelet down- and up-sampling and frequency-aware skip connections carry high-frequency information across resolutions; a progressive input pyramid injects downsampled copies of the conditioned input. The model has no self-attention layers. Its input width is 17 for concatenation and 8 for FiLM, and both variants predict the same eight clean target subbands.

3.5 Healthy-Tissue Objective

The primary experiments derive a shared scale from the voided image: $a = \max(x_v)$. Both the target and condition are divided by a, clipped to $[0, 1]$, and mapped to $[-1, 1]$. We also train one matched sensitivity arm with robust bounds derived from the voided image, $l = \max(0, Q_{0.5}(x_v))$ and $u = Q_{99.5}(x_v)$. In that arm, both volumes are mapped by $(x - l)/(u - l)$ before clipping and mapping to $[-1, 1]$. Degenerate percentile bounds fall back to the max-based rule. Both policies use only inference-available information, and their bounds are retained to restore native intensities after sampling.

Let $e_{k,p} = (\hat{z}_0^{(k)}(p) - z_0^{(k)}(p))^2$ be the squared error in subband k at wavelet-grid position p, and let Ω be the set of grid positions. We set $u_p = 0$ whenever the corresponding $2 \times 2 \times 2$ voxel cell intersects the real tumor mask m_u, and $u_p = 1$ otherwise. The base loss is

$$\mathcal{L}_{\text{wav}} = \frac{1}{8|\Omega|} \sum_{k=1}^{8} \sum_{p \in \Omega} u_p e_{k,p}. \tag{7}$$

For the region-focused term, a cell is active when it intersects m_h; this binary coverage map is Gaussian-smoothed with $\sigma = 2$ on the wavelet grid and multiplied by u to obtain nonnegative weights w_p. We normalize by active mask mass:

$$\mathcal{L}_{\text{mask}} = \frac{\sum_{k=1}^{8} \sum_{p \in \Omega} w_p e_{k,p}}{8 \sum_{p \in \Omega} w_p}, \qquad \mathcal{L} = \mathcal{L}_{\text{wav}} + 10\mathcal{L}_{\text{mask}}. \tag{8}$$

The losses are averaged over the batch. Applying w once to squared error and normalizing by its mass makes the coefficient 10 independent of hole volume and avoids squaring the soft mask weights. Both terms are clamped to zero in cells that touch m_u, so the model is never trained to reproduce tumor as healthy tissue.

3.6 Online Healthy-Mask Augmentation

The fixed-mask setting uses the healthy hole m_h supplied with each training case. For online augmentation, we instead generate a healthy mask m'_h that does not overlap the case's real tumor m_u, void the complete T1n image over $m'_h \cup m_u$, and supervise m'_h while still excluding m_u. A training-only bank of connected tumor components supplies realistic shapes without allowing masks from held-out cases to enter training.

We compare two augmentation policies, each producing five samples per training case. *Random augmentation* uses transformed tumor components with challenge-like inverse size-bin sampling and fixed deterministic samples. *Weighted augmentation* draws 80% tumor-derived components, 15% irregular blobs, and 5% ellipsoids with empirical size sampling, and varies samples across epochs. Both policies enforce a single connected component, at least 800 voxels, at most 25% background overlap, and no overlap with the real tumor. If weighted augmentation exhausts its placement attempts, it falls back to the provided fixed healthy mask; the random arm fails rather than silently changing its distribution.

Figure 2 visualizes the resulting policies using the production sampler. The displayed training case and masks are selected deterministically from mask geometry alone; no reconstruction metric enters the selection. Exact case IDs, seeds, mask hashes, configurations, and the sampled distribution are stored with the figure manifest. Each overlay uses its mask's maximum-area axial slice. Weighted family exemplars are located deterministically across epochs, whereas the distribution audit is fixed at epoch 0.

3.7 Inference and Compositing

At inference, the condition is computed once, Gaussian noise is drawn on the 8×128^3 wavelet grid, and the full 1000-step ancestral sampler is run with EMA weights. Predicted clean coefficients are constrained by inverse-transforming, clamping the voxel-space estimate to $[-1, 1]$, and transforming back. The final coefficients are inverted to 256^3, rescaled by the stored per-case intensity factor, cropped to the native shape, and returned to the original voxel orientation. We then hard-blend the generated image $\hat{x}_{\text{gen}}$ with the observation:

$$\hat{x} = x_v \odot (1 - m) + \hat{x}_{\text{gen}} \odot m. \tag{9}$$

Consequently, every voxel outside the requested inpainting region is preserved exactly.

We additionally evaluate a single-sample reference ($N = 1$) and test-time stochastic reconstruction ensembles with $N \in \{3, 5\}$ trajectories from the same checkpoint and conditioning inputs, using distinct deterministic seeds for each

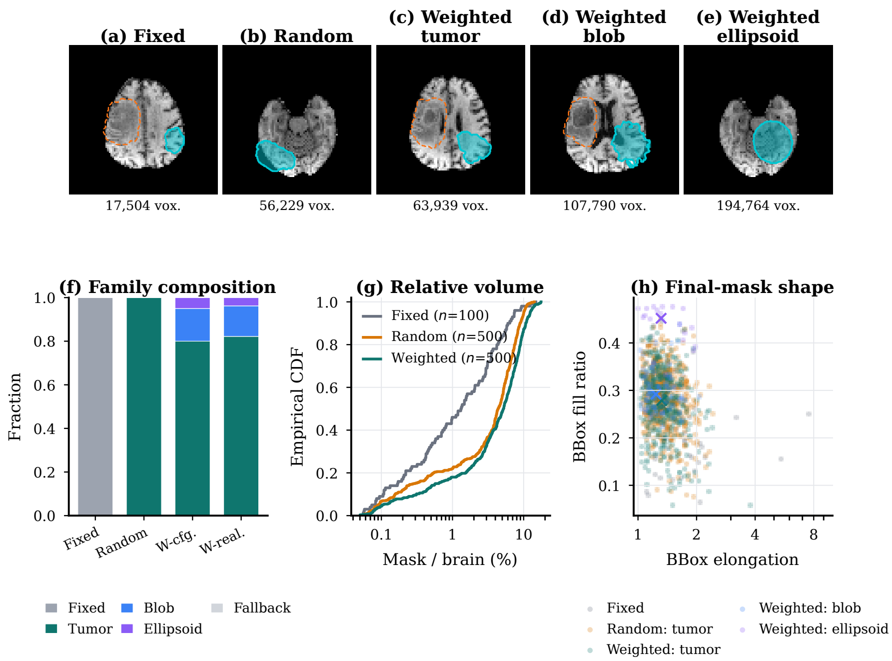

Fig. 2. Healthy-mask policies used for training. (a)–(e) The provided fixed healthy mask, challenge-like random augmentation, and the three families in the weighted policy are overlaid on one audited training case selected nearest the audit median fixed mask-to-brain ratio. Each mask is shown on its own maximum-area axial slice. Cyan denotes healthy supervision; dashed orange denotes the real tumor region excluded from supervision. Examples are mask-geometry representatives from deterministic production samples. (f) Configured and realized family composition. (g) Empirical distributions of final healthy-mask volume relative to brain volume. (h) Final-mask geometry; crosses mark category medians. The descriptive audit uses a deterministic sample of 100 of the 1,151 optimization cases, all five samples per online policy, seed 0, and epoch 0; the component bank excludes the 100-case hold-out. Random masks are epoch-invariant, whereas weighted masks vary by epoch. This figure characterizes sampling and does not by itself establish a reconstruction benefit.

case. Every trajectory is independently decoded, restored to native image space, and hard-composited as above. We aggregate the first N reconstructions voxel-wise by either the arithmetic mean or median and then reapply hard compositing to guarantee exact preservation outside m. The ensemble sizes share a nested sample prefix, so all variants are compared using the same underlying reconstructions. This procedure ensembles samples from one model; it does not combine separately trained networks.

4 Experiments

4.1 Dataset and Held-Out Split

We use the ASNR-MICCAI BraTS Local Synthesis of Healthy Brain Tissue via Inpainting dataset [8,11,2,1], which underlies the BraTS 2026 inpainting task. The public release contains 1,251 training cases and 219 validation cases. A training case provides the complete T1n image, the voided T1n image, the full inpainting mask, an artificially removed healthy sub-mask, and an enlarged tumor sub-mask. The complete T1n image still contains the tumor; it is a valid healthy target only outside the tumor sub-mask. Public validation cases omit the complete image and sub-masks, so their ground truth cannot be scored locally.

We draw a deterministic 100-case hold-out from the training cases with seed 2026, leaving 1,151 cases for optimization. The checked-in case manifest is shared by every arm, and its case IDs are excluded both from the training loader and from the mask component bank. Within this hold-out, we fix a 25-case development subset for checkpoint and ensemble selection. It contains the five cases used for periodic training diagnostics and 20 additional cases chosen by the same deterministic hash ordering. The complementary 75 cases form a locked confirmation cohort. We use the 25-case subset for all model-selection decisions, then evaluate the frozen configuration on the 75-case cohort. We also report the complete 100-case aggregate as a secondary, challenge-style summary. These cohorts come from the released training data and are not independent challenge test sets.

4.2 Preprocessing

Volumes are reoriented to canonical RAS orientation and center-padded from $240 \times 240 \times 155$ to 256^3 without resampling. The inverse crop and original orientation are restored for saved predictions. The three primary concat models use the voided-image maximum for input scaling. A matched weighted-augmentation sensitivity arm instead uses the 0.5th and 99.5th percentiles of the voided image, as defined in Section 3.5. The model then operates on the single-level 3D Haar representation described in Section 3.2. Training-time normalization is distinct from metric normalization: the official evaluator applies its percentile rule to every model output.

Table 1. Final seed-0 concat experiments. "Fixed" denotes the healthy mask supplied with each case. Percentile normalization is evaluated only as a matched sensitivity arm for the weighted policy.

Run	Healthy-mask policy	Input normalization	Role
Concat-fixed	Fixed	Voided maximum	Primary
Concat-random	Random	Voided maximum	Primary
Concat-weighted	Weighted	Voided maximum	Primary
Concat-weighted-percentile	Weighted	0.5–99.5 percentile	Sensitivity

4.3 Experimental Design

Table 1 lists the four final seed-0 concat runs. The three primary arms compare the provided fixed masks with random and weighted online mask augmentation under the original max-based normalization. The fourth arm changes only the normalization of the weighted model and is treated as a sensitivity analysis, not as a full factorial normalization experiment. All four runs use 17-channel concatenation, the same signed mask representation, objective, held-out split, optimizer, and diffusion configuration. Spatial input FiLM remains an exploratory conditioning branch from development; final checkpoint selection is restricted to the concat runs.

4.4 Implementation Details

The U-Net has base width 64, channel multipliers $(1, 2, 2, 4, 4, 4)$, two residual blocks per level, 32-group normalization, SiLU activations, and no attention. We train in FP32 with batch size one per GPU and uniform timestep sampling. AdamW [9] uses initial learning rate 10^{-5}, zero weight decay, and a 300,000-step linear-decay horizon for every final concat run. Training may be resumed across multiple scheduler jobs without changing that horizon. The mask coefficient is 10. An exponential moving average (EMA) with decay 0.9999 is updated after every optimizer step and used for evaluation. Each ablation trains on one NVIDIA A100 40GB GPU using PyTorch 2.2. Checkpoints are saved every 5,000 steps.

During training, the fixed-mask arm samples the same deterministic five held-out cases every 5,000 steps, while the augmentation arms do so every 10,000 steps. These small-cohort curves are used only to monitor training and are not used as final ablation results.

4.5 Checkpoint and Ensemble Selection

Checkpoint screening uses one deterministic reconstruction per case on the 25-case development subset. The planned candidates are 100k, 125k, 150k, 175k, and 200k for the fixed-mask run, and 150k, 200k, 225k, 250k, 275k, and 300k for the random and weighted runs. The percentile-normalized model is first compared with the max-normalized weighted model at the matched 100k checkpoint. Any unavailable candidate is recorded as missing rather than replaced by a nearby step.

For each development case, candidates are ranked separately by SSIM (descending), PSNR (descending), and MSE (ascending). We average the three ranks over cases and select the candidate with the lowest mean rank. Ensemble selection uses the terminal EMA checkpoint at step 299,999 from each of the three primary max-normalized arms. We compare $N \in \{1, 3, 5\}$ seeded reconstructions with voxel-wise mean and median aggregation; the identical mean and median $N = 1$ outputs are treated as one candidate. Ranks are computed within each arm and pooled with equal weight over the three arms, 25 cases, and three metrics to select one global inference rule. This procedure selected five trajectories with voxel-wise mean aggregation. We freeze that rule before evaluating any arm on the 75-case confirmation cohort. Full local evaluation uses EMA weights and the 1000-step ancestral sampler. Predictions are saved as compressed NIfTI volumes with their source geometry, while per-case metrics and aggregate summaries are stored as CSV and JSON.

4.6 Evaluation Metrics

We use the organizers' BraTS local-synthesis implementation [8]. Prediction and target are normalized to $[0, 1]$ using the 0.5th and 99.5th percentiles of the voided image, and metrics are restricted to the provided healthy mask. We report the three ranking metrics: structural similarity (SSIM; higher is better), peak signal-to-noise ratio (PSNR; higher is better), and mean squared error (MSE; lower is better). For the 100-case hold-out we report the per-case mean and sample standard deviation. RMSE, MAE, and MSLE are retained in the machine-readable evaluation output as secondary diagnostics.

5 Results

The final seed-0 training runs and locked confirmation evaluations were still in progress when this manuscript revision was prepared. We therefore do not promote the five-case monitoring curves to quantitative results. Those curves are intended for training diagnostics only and are especially unsuitable for comparison with earlier runs, which used a different mask-loss normalization and a different five-case monitoring cohort.

To make the development history auditable while the final evaluations are pending, Fig. 3 records four corrected-run histories, including the exploratory FiLM branch. The figure uses raw validation events without dashboard smoothing and is a training diagnostic rather than evidence for the final method ranking.

The checkpoint figure will show SSIM, PSNR, MSE, and mean rank on the fixed 25-case development subset. Neither selection figure uses dashboard smoothing.

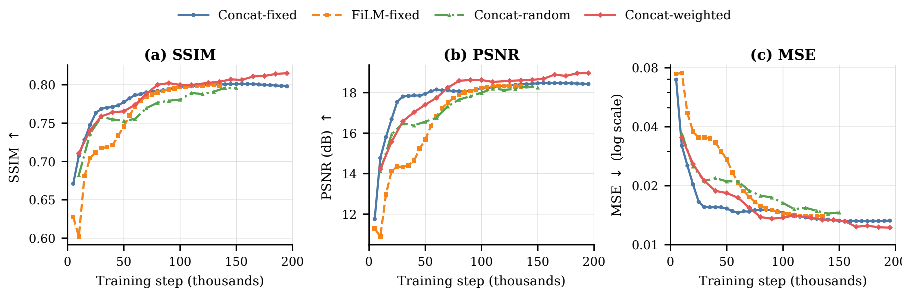

Fig. 3. Interim EMA diagnostics for four seed-0 development runs on the same deterministic five-case monitoring cohort. The snapshot contains concat fixed, concat random, concat weighted, and exploratory FiLM weighted runs. Fixed-mask monitoring occurs every 5,000 steps and augmented-run monitoring every 10,000 steps. Curves end at the latest available event and are not used for checkpoint or method selection. Final claims use the predeclared development and confirmation cohorts.

5.1 Development-Set Ensemble Selection

Table 2 reports the preconfirmation ensemble screen on the 25-case development subset. For the fixed-mask arm, ensembling had no material effect on the aggregate metrics: over all five candidates, SSIM spanned only 0.709715–0.709722, PSNR 18.178827–18.179194 dB, and MSE 0.015027–0.015028 at the reported precision. Although distinct seeds changed the reconstruction inside the scored healthy region for all 25 cases, averaging did not consistently reduce the fixed arm's error.

The random and weighted augmentation arms behaved differently. Mean aggregation with $N = 5$ achieved the lowest joint rank for both arms, whereas $N = 1$ ranked last. The fixed arm alone slightly preferred mean aggregation with $N = 3$, but the predeclared global selection pools all three arms. The pooled rank therefore selects $N = 5$ with voxel-wise mean aggregation, which is frozen for the 75-case confirmation evaluation.

Table 2. Challenge-style mean rank on the 25-case development subset for the terminal EMA checkpoints. Ranks are averaged over SSIM, PSNR, and MSE; lower is better. The $N = 1$ mean/median duplicate is shown once. Bold values mark the best configuration within each column.

Configuration	Fixed	Random	Weighted	Pooled
$N = 1$	2.9200	4.7067	4.4000	4.0089
Mean, $N = 3$	**2.7867**	2.1333	2.6933	2.5378
Median, $N = 3$	2.9733	3.5333	3.7733	3.4267
Mean, $N = 5$	3.2400	**1.6000**	**1.5867**	**2.1422**
Median, $N = 5$	3.0800	3.0267	2.5467	2.8844

The final quantitative table will be populated from the immutable aggregate JSON and per-case CSV files produced by the shared evaluator. Its primary columns will report mean $\pm$ sample standard deviation for SSIM, PSNR, and MSE on the locked 75-case confirmation cohort. A secondary 100-case aggregate will support comparison with challenge-style local evaluations. Autoencoder and Pix2Pix baselines will be scored on the same case manifests with the same official evaluator. The official validation submission will be reported separately because its ground truth is hidden.

The qualitative figure will select the cases nearest the 10th, 50th, and 90th percentiles of confirmation-cohort SSIM under the frozen method. Each row will show the voided input, one-sample reconstruction, selected-ensemble reconstruction, ground truth, and absolute-error maps. Case selection, slice selection, intensity windows, and metric labels are deterministic and saved with the figure manifest.

6 Discussion

The design separates three questions that are otherwise easy to conflate: how the mask is represented, how the condition enters the denoiser, and how healthy holes are sampled during training. The one-channel signed mask preserves explicit inside/outside semantics without expanding the mask into eight Haar subbands. Concatenation exposes all 17 channels directly to the U-Net. Input FiLM reduces the main U-Net input to eight channels and begins as an identity transform, but it is not free: its conditioning network applies two 32-channel 3D convolutions at the full 128^3 wavelet resolution. The ablation must therefore be judged on reconstruction quality and measured runtime, not on U-Net input width alone.

The corrected region-focused loss has a stable interpretation across hole sizes: its coefficient multiplies an average in-mask squared error. Excluding every wavelet cell that touches the real tumor prevents direct regression toward pathological appearance, although this conservative rule also removes some healthy boundary voxels from supervision. Online mask augmentation increases the diversity of supervised healthy holes while retaining the real tumor in the excluded region. Tumor-derived shapes are anatomically plausible masks, whereas blobs and ellipsoids broaden shape coverage at the risk of departing from the challenge distribution.

The development-set ensemble response also differed by mask policy. Multiple trajectories changed the scored region for every fixed-arm case, yet averaging did not materially change its aggregate metrics. A plausible interpretation is that the fixed model's remaining error is predominantly systematic across trajectories, which an ensemble cannot remove. Random and weighted mask augmentation may instead retain more seed-dependent reconstruction error, allowing a mean ensemble to cancel part of that variation. This is a development-set observation rather than a causal result; the frozen $N = 5$ mean rule is evaluated without retuning on the confirmation cohort.

Several limitations remain. Only the T1n modality is used, each training arm has one seed, and the local 100-case hold-out comes from the released training cohort rather than an independent test set. The 75-case cohort is therefore a locked internal confirmation set, not an external test set. The percentile experiment covers only the weighted mask policy, so it measures one matched normalization contrast rather than a normalization-by-augmentation interaction. Spatial FiLM was explored during development but was not carried through the final checkpoint-selection protocol. Finally, full 1000-step ancestral sampling is computationally expensive, especially for multiple reconstructions. Accelerated sampling, multimodal conditioning, more training seeds, and external evaluation remain future work. Until the saved confirmation evaluations are available, no final ranking should be inferred from the five-case monitoring cohort.

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References

1. Baid, U., Ghodasara, S., Mohan, S., Bilello, M., Calabrese, E., Colak, E., Farahani, K., Kalpathy-Cramer, J., Kitamura, F.C., Pati, S., et al.: The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification. arXiv preprint arXiv:2107.02314 (2021)
2. Bakas, S., Akbari, H., Sotiras, A., Bilello, M., Rozycki, M., Kirby, J.S., Freymann, J.B., Farahani, K., Davatzikos, C.: Advancing the cancer genome atlas glioma MRI collections with expert segmentation labels and radiomic features. Scientific Data **4**(1), 1–13 (2017)
3. Dhariwal, P., Nichol, A.Q.: Diffusion models beat GANs on image synthesis. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 34, pp. 8780–8794 (2021)
4. Durrer, A., Wolleb, J., Bieder, F., Friedrich, P., Cattin, P.C.: Denoising diffusion models for 3d healthy brain tissue inpainting. In: MICCAI Workshop on Deep Generative Models. Springer (2024)
5. Friedrich, P., Durrer, A., Wolleb, J., Cattin, P.C.: cWDM: Conditional wavelet diffusion models for cross-modality 3d medical image synthesis. In: MICCAI Brain Tumor Segmentation (BraTS) Challenge (2024), arXiv:2411.17203
6. Friedrich, P., Wolleb, J., Bieder, F., Durrer, A., Cattin, P.C.: WDM: 3d wavelet diffusion models for high-resolution medical image synthesis. In: MICCAI Workshop on Deep Generative Models. Springer (2024)
7. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33, pp. 6840–6851 (2020)
8. Kofler, F., Meissen, F., Steinbauer, F., Graf, R., Oswald, E., de la Rosa, E., et al.: The brain tumor segmentation (BraTS) challenge 2023: Local synthesis of healthy brain tissue via inpainting. arXiv preprint arXiv:2305.08992 (2023)
9. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (ICLR) (2019)

10. Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., Van Gool, L.: RePaint: Inpainting using denoising diffusion probabilistic models. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 11461–11471 (2022)
11. Menze, B.H., Jakab, A., Bauer, S., Kalpathy-Cramer, J., Farahani, K., Kirby, J., et al.: The multimodal brain tumor image segmentation benchmark (BRATS). IEEE Transactions on Medical Imaging **34**(10), 1993–2024 (2015)
12. Nichol, A.Q., Dhariwal, P.: Improved denoising diffusion probabilistic models. In: International Conference on Machine Learning (ICML). pp. 8162–8171. PMLR (2021)
13. Perez, E., Strub, F., de Vries, H., Dumoulin, V., Courville, A.: FiLM: Visual reasoning with a general conditioning layer. In: AAAI Conference on Artificial Intelligence. vol. 32 (2018)
14. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 10684–10695 (2022)
15. Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention (MICCAI). pp. 234–241. Springer (2015)
16. Saharia, C., Chan, W., Chang, H., Lee, C., Ho, J., Salimans, T., Fleet, D., Norouzi, M.: Palette: Image-to-image diffusion models. In: ACM SIGGRAPH 2022 Conference Proceedings. pp. 1–10 (2022)
17. Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models. In: International Conference on Learning Representations (ICLR) (2021)
18. Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., Poole, B.: Score-based generative modeling through stochastic differential equations. In: International Conference on Learning Representations (ICLR) (2021)